

AIRLINE LOYALTY

BUSINESS INTELLIGENCE SOLUTION

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1. Data Overview

The project is based on a comprehensive Airline Loyalty Dataset. The dataset is used to build a full-stack Business Intelligence solution using SSIS for ETL, a Tabular model for analytics, and Power BI for interactive reporting.

Data Source

- Input Format: Excel file
- Tables:
 - Flight Activity Table: Loyalty_Number, Year, Month, Total_Flights, Distance, Points_Accumulated, Points_Redeemed, Dollar_Cost_Points_Redeemed
 - Customer Loyalty History Table: Loyalty_Number, Country, Province, City, Postal Code, Gender, Education, Salary, Marital Status, Loyalty_Card, CLV, Enrollment_Type, Enrollment_Year, Enrollment_Month, Cancellation_Year, Cancellation_Month

Flight Activity Attributes

Attribute	Description
Loyalty_Number	Unique identifier assigned to each loyalty member. Used as the primary key and join key between tables.
Year	Calendar year of flight activity. Used for year-over-year trend analysis.
Month	Month of flight activity (1-12). Used for seasonal and monthly analysis.
Total_Flights	Total number of flights taken by the member in that month/year period.
Distance	Total distance flown in miles or kilometers during the period.
Points_Accumulated	Loyalty points earned from flights in the given period.
Points_Redeemed	Loyalty points redeemed by the member during the period.
Dollar_Cost_Points_Redeemed	Monetary value (in dollars) of the points redeemed during the period.

Customer Loyalty History Attributes

Identification & Classification Attributes

Attribute	Description
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Attribute	Description
Loyalty_Number	Unique identifier for each customer. Primary key linking to flight activity.
Gender	Gender of the loyalty member (Male / Female).
Education	Highest level of education attained (e.g., Bachelor, College, High School or Below).
Marital Status	Marital status of the member: Single, Married, or Divorced.
Loyalty_Card	Tier of loyalty card held: Star, Nova, or Aurora.
Enrollment_Type	Type of enrollment: Standard or 2018 Promotion.

Geographic Attributes

Attribute	Description
Country	Country of residence of the loyalty member.
Province	Province or state of residence.
City	City of residence.
Postal Code	Postal or ZIP code of the member's address.

Financial & Lifetime Value Attributes

Attribute	Description
Salary	Annual salary of the member (in dollars). Used for customer segmentation analysis.
CLV	Customer Lifetime Value — estimated total revenue from the member over their relationship with the airline.

Enrollment & Cancellation Attributes

Attribute	Description
Enrollment_Year	Year the member enrolled in the loyalty program.
Enrollment_Month	Month the member enrolled in the loyalty program.
Cancellation_Year	Year the member cancelled their loyalty membership (NULL if still active).
Cancellation_Month	Month the member cancelled their loyalty membership (NULL if still active).

2. Conceptual & Logical & Physical Model

This solution implements a star schema design in the Data Warehouse layer for high performance and ease of analysis.

2.1 Conceptual Model

The conceptual model defines the core entities, their roles, and relationships within the business context — abstracted away from technical implementation. It describes what data exists and how it is related, without getting into physical storage or data types.

The conceptual model consists of one central fact entity and seven surrounding dimension entities, forming a classic star schema structure.

Entities (Blocks):

- `dbo.fact_table` (Fact Table)
- `dbo.dim_customer` (Dimension)
- `dbo.dim_location` (Dimension)
- `dbo.dim_gender` (Dimension)
- `dbo.dim_education` (Dimension)
- `dbo.dim_enrollment` (Dimension)
- `dbo.dim_cancel` (Dimension)
- `dbo.dim_status` (Dimension)

All dimensions are related to the fact table in a 1:M relationship.

2.2 Logical Model

The logical model is defined using the following entities:

Dimensions:

- `dim_customer`: Customer metadata including `loyalty_number`, `loyalty_card`
- `dim_location`: Geographic data including `id_location`, `country`, `province`, `city`, `postal_code`
- `dim_gender`: `id_gender`, `gender`
- `dim_education`: `id_education`, `education`
- `dim_enrollment`: `id_enrollment`, `enrollment_type`, `enrollment_year`, `enrollment_month`
- `dim_cancel`: `id_cancel`, `cancellation_year`, `cancellation_month`
- `dim_status`: `id_status`, `marital_status`

Fact Table:

- `fact_table`: Fact table referencing all dimensions and housing measurable flight metrics (total_flights, , salary, CLV, distance, points_accumulated, points_redeemed, dollar_cost_points_redeemed).

Each dimension is designed to support one-to-many relationships with the fact table.

2.3 Physical Model

The physical model implements the logical star schema within a SQL Server-based Data Warehouse. It focuses on how data is stored in the database, table structures, and physical organization — not on how data gets there.

Databases Involved:

- `DWH_AirlineLoyalty` → Final Data Warehouse containing structured data for analytics

Tables Stored Physically:

- `fact_table` (Fact table storing measurable flight and loyalty point metrics)
- `dim_customer` (Customer descriptive details, loyalty card tier, and CLV)
- `dim_location` (Geographic hierarchy from city to country)
- `dim_gender` (Gender lookup dimension)
- `dim_education` (Education level lookup dimension)
- `dim_enrollment` (Enrollment type and date dimension)
- `dim_cancel` (Cancellation year and month dimension)
- `dim_status` (Marital status lookup dimension)

Structure:

- Surrogate keys used for all dimension tables
- Foreign key relationships enforced between `fact_table` and all dimension tables
- Star schema optimized for analytical querying
- All text fields normalized and typecast to avoid BLOB issues in reporting tools

3. SSIS Overview

The SSIS (SQL Server Integration Services) layer orchestrates the ETL (Extract, Transform, Load) process in three clearly defined stages, aligned with the architecture's layered design:

ETL Database Layers:

- ODS_AirlineLoyalty – Raw operational data store
- STG_AirlineLoyalty – Cleaned and conformed staging layer
- DWH_AirlineLoyalty – Final dimensional warehouse

A. ODS Package – Raw Data Load

Objective: Ingest raw Airline Loyalty Excel data and apply basic cleaning and conversions before loading into the ODS layer.

ETL Flow:

- Source: Excel file with Flight Activity and Customer Loyalty History sheets
- Destination: ODS_AirlineLoyalty database

Key Tasks:

- Truncate existing data (Execute SQL Task)
- Convert all date/year/month fields to nvarchar at start
- Load both sheets as separate data flows

SSIS Components Used:

- Excel Source
- Derived Column
- Data Flow Task
- OLE DB Destination

B. STG Package – Staging and Transformation

Objective: Prepare and organize cleaned data into dimension staging tables for further warehousing.

ETL Flow:

- Source: ODS_AirlineLoyalty database
- Destination: STG_AirlineLoyalty database

Key Tasks:

- Truncate STG dimension tables
- Store and structure data into:
 - dim_customer → with validation of loyalty_card, enrollment_type and all foreign keys
 - dim_location → validated country, province, city, postal_code
 - dim_gender → distinct gender values extracted and deduplicated
 - dim_education → distinct education levels extracted and validated
 - dim_enrollment → enrollment_type, enrollment_year, enrollment_month validated
 - dim_cancel → cancellation_year, cancellation_month validated (NULL handling for active members)
 - dim_status → distinct marital status values extracted

SSIS Components Used:

- Sequence Container
- Data Flow Tasks
- Derived Column
- Conditional Split (active vs. cancelled members)
- OLE DB Destination

C. DWH Package – Warehouse Load & Lookup

Objective: Populate the dimensional model in DWH_AirlineLoyalty with clean, fully integrated data.

ETL Flow:

- Source: STG_AirlineLoyalty database
- Destination: DWH_AirlineLoyalty database

Key Tasks:

- Truncate final dimension and fact tables
- Load all dimensions: dim_customer, dim_location, dim_gender, dim_education, dim_enrollment, dim_cancel, dim_status
- Populate fact_table with:
 - Lookups: Customer, Location, Gender, Education, Enrollment, Cancel, and Status surrogate keys
 - Data Type Enforcement: Ensures all fields are compatible with Power BI (avoids BLOB output)

- Derived metrics: net points balance (Points_Accumulated - Points_Redeemed)

SSIS Components Used:

- Two Sequence Containers (Dimensions / Fact)
- Lookup
- Derived Column
- Data Flow Tasks
- OLE DB Destination